

```
create Database Online_Learning
```

```
use Online_Learning
```

```
CREATE TABLE Instructors (  
    InstructorID INT PRIMARY KEY,  
    FullName VARCHAR(100),  
    Email VARCHAR(100),  
    JoinDate DATE  
);
```

```
CREATE TABLE Categories (  
    CategoryID INT PRIMARY KEY,  
    CategoryName VARCHAR(50)  
);
```

```
CREATE TABLE Courses (  
    CourseID INT PRIMARY KEY,  
    Title VARCHAR(100),  
    InstructorID INT,  
    CategoryID INT,  
    Price DECIMAL(6,2),  
    PublishDate DATE,  
    FOREIGN KEY (InstructorID) REFERENCES Instructors(InstructorID),  
    FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID)  
);
```

```
CREATE TABLE Students (  
    StudentID INT PRIMARY KEY,  
    FullName VARCHAR(100),  
    Email VARCHAR(100),  
    JoinDate DATE  
);
```

```
CREATE TABLE Enrollments (  
    EnrollmentID INT PRIMARY KEY,  
    StudentID INT,  
    CourseID INT,  
    EnrollDate DATE,  
    CompletionPercent INT,  
    Rating INT CHECK (Rating BETWEEN 1 AND 5),  
    FOREIGN KEY (StudentID) REFERENCES Students(StudentID),  
    FOREIGN KEY (CourseID) REFERENCES Courses(CourseID)  
);
```

```
-- Instructors
```

```
select * From Instructors
```

```
INSERT INTO Instructors VALUES
```

```
(1, 'Sarah Ahmed', 'sarah@learnhub.com', '2023-01-10'),
```

```
(2, 'Mohammed Al-Busaidi', 'mo@learnhub.com', '2023-05-21');
```

-- Categories

select * From Categories

INSERT INTO Categories VALUES

(1, 'Web Development'),

(2, 'Data Science'),

(3, 'Business');

-- Courses

select * From Courses

INSERT INTO Courses VALUES

(101, 'HTML & CSS Basics', 1, 1, 29.99, '2023-02-01'),

(102, 'Python for Data Analysis', 2, 2, 49.99, '2023-03-15'),

(103, 'Excel for Business', 2, 3, 19.99, '2023-04-10'),

(104, 'JavaScript Advanced', 1, 1, 39.99, '2023-05-01');

-- Students

select * From Students

INSERT INTO Students VALUES

(201, 'Ali Salim', 'ali@student.com', '2023-04-01'),

(202, 'Layla Nasser', 'layla@student.com', '2023-04-05'),

(203, 'Ahmed Said', 'ahmed@student.com', '2023-04-10');

-- Enrollments

select * From Enrollments

INSERT INTO Enrollments VALUES

(1, 201, 101, '2023-04-10', 100, 5),

(2, 202, 102, '2023-04-15', 80, 4),

(3, 203, 101, '2023-04-20', 90, 4),

(4, 201, 102, '2023-04-22', 50, 3),

(5, 202, 103, '2023-04-25', 70, 4),

(6, 203, 104, '2023-04-28', 30, 2),

(7, 201, 104, '2023-05-01', 60, 3);

--Beginner Level

select Count(StudentID)

From Students

select Count(EnrollmentID)

From Enrollments

select (CourseID),AVG(Rating) as avgRating

From Enrollments

Group by CourseID

```
select ( FullName) ,count( *) as total_courses
```

```
From Instructors
```

```
Group By FullName
```

```
select ( Title), count ( *) no_courses
```

```
From Courses
```

```
Group by Title
```

```
select ( CategoryID ), count ( *) no_courses
```

```
From Courses
```

```
Group by CategoryID
```

```
select CourseID, count ( StudentID) as no_students
```

```
From Enrollments
```

```
Group by CourseID
```

```
select CategoryID, avg ( Price) as avgPrice
```

```
From Courses
```

```
Group by CategoryID
```

```
select max( Price) as maxPrice
```

```
From Courses
```

```
select CourseID , min( Rating) as min_Rating, max ( Rating) as max_Rating, avg ( Rating) as  
avg_Rating
```

```
From Enrollments
```

```
Group by CourseID
```

```
select count ( *) as num_students
```

```
From Enrollments
```

```
where rating = 5
```

```
select StudentID, count ( *) as num_students
```

```
From Enrollments
```

```
where rating = 5
```

```
Group by StudentID
```